

BOMI KWON

Seoul, Republic of Korea · bomi9970@sookmyung.ac.kr · kwonbomi.github.io

Research Interests: Diffusion-based structured generation and controllable generative modeling.

EDUCATION	Sookmyung Women's University	Seoul, Republic of Korea
	Master of Science, Department of Computer Science GPA: 4.41/4.5	Mar. 2025 – Feb. 2027 (expected)
	Bachelor of Science, Department of Statistics	Mar. 2020 – Feb. 2025
	Bachelor of Business Administration, Department of Business Administration	
PUBLICATIONS & PRESENTATIONS	B. Kwon and K. Y. Lee (2026). <i>Text-Conditioned Scene Graph Sequence Generation via Discrete Diffusion</i> . Proceedings of the 35th ACM International Conference on Information and Knowledge Management (CIKM '26). Accepted Short Paper.	
	B. Kwon and K. Y. Lee (2026). <i>Text-driven 2D Human Pose Generation via Diffusion in Directional Angle Space</i> . The Transactions of the Korea Information Processing Society, 15(7), 610–619.	
	B. Kwon , S. Park, Y. Park, and K. Y. Lee (2026). <i>Text-Conditioned Diffusion for Static Humanoid Robot Pose Generation with Joint-Limit Embedding and Self-Collision Guidance</i> . Submitted to KICS 2026.	
	B. Kwon and K. Y. Lee (2026). <i>Diffusion-based Skeleton Pose Editing</i> . 2026 Spring Conference of the Korean Institute of Smart Media and the Korean Society of Digital Industry.	
	B. Kwon and K. Y. Lee (2025). <i>A Graph Generation Model for Text-based Graph Editing</i> . Korea Software Congress 2025 (KSC 2025).	
	B. Kwon and K. Y. Lee (2025). <i>Development of a Large-Scale Graph Retrieval System based on LLM and Graph Database Providing a Natural Language Interface</i> . Korea Computer Congress 2025 (KCC 2025).	
HONORS & AWARDS	Principal Investigator, 2026 STEM Research Team Project	Apr. 2026 – Oct. 2026
	Korea Foundation for Women in Science, Engineering and Technology (WISET) Selected as one of 50 research teams nationwide; awarded KRW 7,000,000 in competitive research funding.	
	Sookmyung Leadership Activity Scholarship	Jan. 2023
RESEARCH EXPERIENCE	Sookmyung Women's University	Seoul, Republic of Korea
	Graduate Researcher, Data Intelligence Laboratory	Mar. 2025 – Present
	Advisor: Prof. Ki Yong Lee	
	WISET	
	Principal Investigator, STEM Research Team Project	Apr. 2026 – Oct. 2026
	- Lead a team of two undergraduate researchers developing synthetic human-pose generation methods for industrial safety. - Manage budget execution and reporting for the funded research project.	
PROJECTS	Text-Conditioned Scene Graph Sequence Generation via Discrete Diffusion	2025 – 2026
	- Developed a discrete diffusion framework for generating time-ordered scene graph sequences from natural-language descriptions of human-object interactions. - Designed a Transformer-based model for modeling within-frame object relations and temporal transitions; constructed training data from Charades-STA and Action Genome and evaluated against retrieval and generative baselines. - Led the project as first author from problem formulation and model development through experiments and manuscript preparation.	
	Text-Conditioned Humanoid Robot Pose Generation with Physical Constraints	2026
	- Developed a text-conditioned diffusion model for generating static poses for the Unitree G1 humanoid under joint-range and self-collision constraints. - Retargeted human motion data to Unitree G1 in MuJoCo and incorporated physical constraints into the generation process while preserving text-based control and pose diversity.	

Text-driven 2D Human Pose Generation via Diffusion in Directional Angle Space

2025 – 2026

- Proposed a directional-angle representation that models body segments by their directions rather than Cartesian joint coordinates.
- Built a BGE-M3-conditioned diffusion model for pose generation while preserving consistent skeletal proportions.

AISUM Industry–Academic Collaboration — Product Matching and Retrieval System

2025 – Present

- Developed product retrieval pipelines spanning visual product matching and semantic recommendation for large-scale product data.
- Evaluated visual embedding models and ranking strategies, and built semantic retrieval using BGE-m3-ko/KURE-v1 embeddings and Qdrant.
- Integrated the retrieval functions into an MCP-based interface and implemented a LangGraph agent workflow for multi-stage product search.

Large-Scale Graph Retrieval System with LLM and Neo4j

2025

- Developed an LLM-based natural-language interface that converts user queries into Cypher for retrieval over large-scale graph data in Neo4j.

fitapat (ZOOC)

Founding Team Member / Product & Marketing Lead

Jul. 2023 – Aug. 2024

- Co-founded and led product development for an AI-based pet customization service that generated personalized characters from user-provided photos and connected them to customized merchandise.
- Led product and go-to-market execution through launch; the platform and physical products were showcased at CES 2024.

SOPT

Vice President, 32nd Cohort

Jan. 2023 – Jul. 2023

Product Manager, 31st Cohort

Jul. 2022 – Jan. 2023

- Led a nationwide university IT and startup community; previously served as Product Manager for ZOOC, a digital service for pet owners.

Sookmyung Admissions Ambassador “Polaris” — Vice President

Mar. 2021 – Feb. 2023

Statistics Korea (KOSTAT) — Volunteer Statistics Educator

May 2021 – Dec. 2021

Amorepacific × Sookmyung A.I. Team Project — Team Project

Oct. 2024 – Dec. 2024

SOOK-TAT — Statistics Data Analysis Society, Member

Sep. 2022 – Jul. 2023

Department of Statistics Student Council — Member

Mar. 2020 – Dec. 2021

KB Campus Star — Student Ambassador

May 2022 – Nov. 2022

Programming: Python, R, SQL

ML & AI: PyTorch, Diffusion Models, Transformers, LangChain, LangGraph, MCP

Databases & Retrieval: Qdrant, PostgreSQL/pgvector, Neo4j

Simulation: MuJoCo

Tools: Git, GitHub

Certification: ADsP